

QUESTION FOR DISCUSSION AND RECAPITULATIONS

1. What are the Meaning, Concept and cycle of Micro Teaching?
2. Explain the skill of introduction and questioning and the component?
3. Difference between Micro-Teaching and Macro Teaching.
4. Explain about Skill of introduction, Skill of explaining, and Skill of reinforcement.
5. Explain about Skill of illustrating with examples, Skill of questioning.
6. Explain about Skill of stimulus variation, Skill of achieving closure.
7. Explain about Skill of probing questioning components.
8. Explain suitable any 3 Skills for your subject.
9. How will you explain you subject in class?
10. Practicing a Link Lesson with Multiple Teaching Skills (For 20 Minutes).
11. Prepare link lesson plan and how to practice that in front of peers in the class for level I.
12. Prepare link lesson plan and how to practice that in front of peers in the class for level II.

UNIT 3 APPROACHES OF TEACHING COMMERCE AND ACCOUNTANCY

3.1 APPROACHES OF LESSON PLANNING

3.1.1 Lesson Plan

After planning for the unit we have to concentrate on each lesson plan. Lesson plan actually is a plan of action. The teacher has to decide her work in a particular period. He has to plan his tools and materials. Lesson plan is a teacher's mental and emotional visualization of class room activities. For successful and effective teaching, lesson plan is very essential. Lesson plan explains to be fulfilled and the teaching methods and aids to be used.

A lesson plan is a blue print, a guide map, a creative piece of work and the chart for classroom teaching in one period of 45 minutes duration. It is a systematic approach for the development of concepts, skills, understanding etc.

Lesson plan is an experience in anticipated teaching outcome of the learner and the procedure for learning process. The vivid imagination of the classroom situation speaks of the classroom experience. For effective anticipated experience, there requisites are:

- a) Mastery of subject matter, the materials and activities.
- b) Through knowledge of the environment forces i.e. previous knowledge and experiments of the learners.
- c) Understanding the psychology of mental process and laws of learning.
- d) Understanding of the teaching techniques, goals, aims, skills, methods etc.

3.1.2 Characteristics of an Effective Lesson Plan

A good lesson plan should:

1. Include clearly mentioned instructional objectives.
2. Include the list of teaching aids, the details about how and when to be used.
3. Include motivational techniques.
4. Include evaluation techniques.
5. Include student centred activities.
6. Based on teaching skills include set induction and closure.
7. Show connection between previous lesson and future lesson.
8. Provide sufficient scope for mixed ability grouping.
9. Anticipate pupil's difficulties in question.
10. Encourage thought provoking and interesting in questions from students.
11. Provide effective illustration thought the period.
12. Include relevant home assignment.
13. Provide opportunity to practise skills.
14. Provide enough opportunity to use new methods of teaching.
15. Provide proper time allocation.
16. Provide for individual differences.
17. Be relevant to actual situation with particular reference to the institution, its philosophy practices syllabus, time table and socio economic, cultural background of the learners.

3.1.3 Bases of Lesson Plan

A lesson plan is based on the following theoretical assumptions.

1. Preliminary knowledge and abilities required.
2. Steps followed in writing lesson plan.
3. Evaluation of lesson plan.

Preliminary knowledge and abilities:

A prospective biology teacher needs an understanding of the preliminary knowledge which the lessons plan in developed. This is essential to plan the following:

1. Identification of teaching points or concepts.
2. Writing specific objects.
3. Developing learning activities.
4. Selecting appropriate teaching strategies.
5. Identifying relevant instructional materials.
6. Writing the test items.

3.1.4 Importance of Lesson-Plan

For constructing a lesson-plan, following elements are important

1. Teacher should have capacity to write teacher-objective in a general or behavioral form.
2. Teacher should have controlling distribution and content.
3. Teacher should have complete knowledge about use of teaching- methods and teaching material and he also should know that how these can be used effectively.
4. Teacher should have capacity to co-ordinate between teaching processes in access models.
5. Teacher should also know how to construct circumstances of inspection and evolutionary questions.
6. Teacher should know general comprehension of model of lesson plan.

3.1.5 Advantages of Lesson Plan

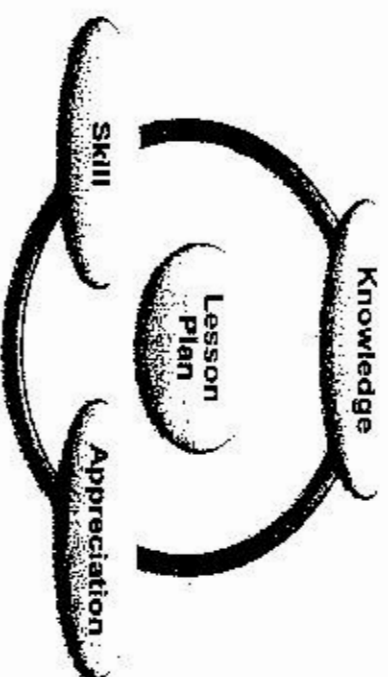
The following are the advantages of planning a lesson.

- ❖ Helps to think about meaningful learning experiences to be provided in the classroom.
- ❖ Helps to identify the specific objectives related to particular lesson. This in turn may help to determine the line of action to be provided in the classroom.
- ❖ Helps student to learn specific concepts, skills and values.
- ❖ Provides an outline of content and activities.
- ❖ Provides opportunity to select and organise activities that may be helpful in achieving the objectives.
- ❖ Eliminates the chance of trial and error in teaching.
- ❖ Helps to select proper content in relation to specific objectives

- ❖ Serves as creative piece of work where a teacher can utilize his knowledge, skills, materials and experience.
- ❖ Provides situation for physical, intellectual and social involvement.
- ❖ Helps teacher to provide opportunity for pupils to satisfy their interests.
- ❖ Helps to visualize the needs to be mixed ability of the pupils.
- ❖ Develops confidence in the science teacher to teach systematically in a planned and organized way.

3.2 TYPES OF LESSON PLAN

The lesson plans may be classified into three categories on the basis of learning objectives



a) Lesson Plan related to knowledge

The knowledge, comprehension, application, analysis and synthesis objectives are achieved by this plan, the main emphasis is given on presentation of content. These type of lessons are prepared in teaching social studies

b) Lesson plans related to Appreciation

This type of lessons are prepared for developing attitudes, feeling and values among students. This type is designed in teaching poetry of English, Hindi, Sanskrit and Regional language

c) Lesson plans related to skill

This skill lessons are designed to achieve the psychomotor learning objectives. It includes skills of reading, writing and

speaking. This types are designed in teaching drawing, music home science, agriculture, technical subjects and craft etc.

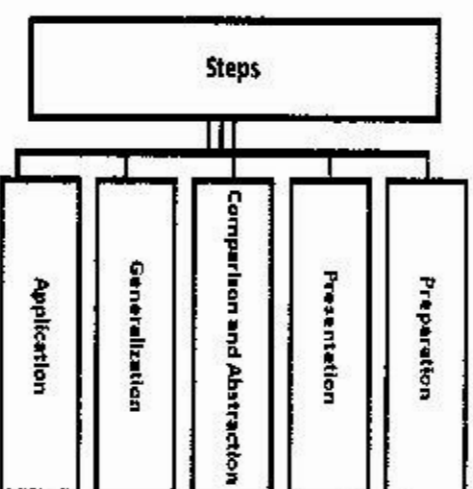
Various Forms of Lesson plans

1. Herbartian approach to lesson planning
2. Bloom's or Evaluation Approach to lesson planning
3. RCEM Approach to lesson planning

3.2.1 Herbartian approach to lesson planning

- ❖ It is an ancient
- ❖ It is used in training institution

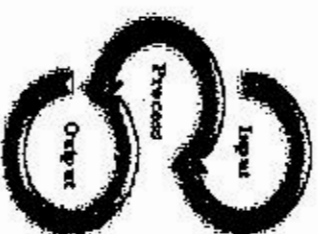
Steps followed in this approach are:



3.2.2 RCEM Approach

The Indian educationists develop this approach. It is a modification of above two approaches.

It consists of three aspects



1) Input:**Expected Behavioral outcomes (EBOs)**

- ❖ Knowledge,
- ❖ Comprehension,
- ❖ Application
- ❖ Creativity

2) Process:

- ❖ Selection of strategies and tactics
- ❖ Interaction of teacher and students

3) Output:

- ❖ Real learning outcomes (RLOs)
- ❖ Changes and modification of behaviours

RCEM (Regional College of Education, Mysore) design of Lesson Plan

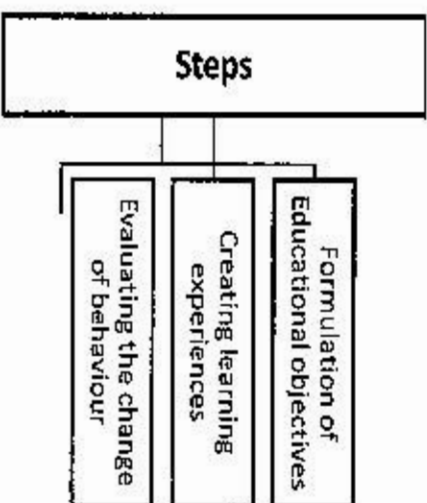
Input	Process		Output
Instruction	Communication Strategy		Evaluation
Expected Behaviour outcomes (EBOS)	Learning Experiences		Real Learning Outcomes (RLOS)
	Teacher Activities	Student Activities	
1. Knowledge objective	Lecture Demonstration Chart Maps Explanation Question and Answer Method	Listening Observation Taking Notes Interaction	Review Questions Define state Describe Name list
2. Understanding Objective	Discussion Problem solving Question and answer method Demonstrated	Participation in group discussion, listening observation initiation	Interpret Translate Explain Discriminate Problem solving

3. Application objectives	Group discussion Laboratory work Problem solving Question and answer method	Experiment, Use of knowledge for his problem	Practical tests Situational test Essay type test Observation
4. Creativity objectives	Individual work Group discussion Problem solving	Analysis of Synthesis elements Establishes new Relationship	Essay type test Oral test Problem solving test observation Situational test

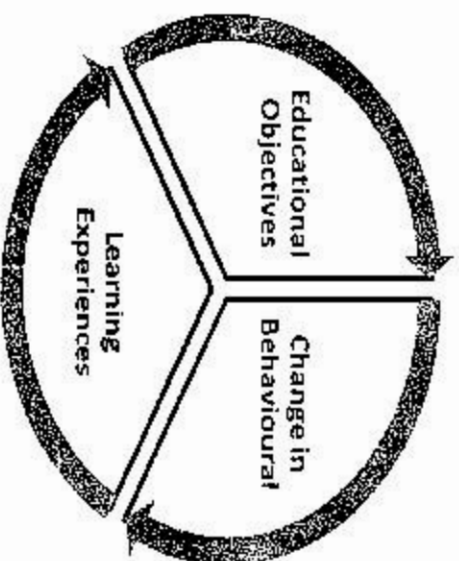
This chapter indicates three aspects: input, process, output can be implemented in organizing teaching. The input aspect concerns with EBOs of teaching, it also includes entering behaviours of the learners. The process aspect involves teacher and student interaction. The teacher and students activities are determined. The output aspect deals with the RLOs. The evaluation devices are specified for measuring the criterion behaviours

3.2.3 Blooms or Evaluation Approach:

It is an innovative approach in the field of education

Steps:

It considers education is the tripolar process:



This table indicates the different type of teaching strategies for achieving different learning objectives

Teaching Objectives		Learning Objectives	
Knowledge		Cognitive	
Understanding		Affective	
Application		Psychomotor	
Creativity		Psychomotor	

Steps in lesson plan:

1. General information:

The general information regarding the subject, unit, less, class, estimated time and date should be provided.

2. Identification of teaching points:

The lesson carefully read. The major and sub concepts are identified to make teaching and learning easily. The teaching points are noted in an organised manner step by step.

3. Writing instructional objectives:

General and specific instructional objectives should be provided. These objectives reflect learning outcomes. So carefully thought should be given in writing Instructional objectives.

4. Instructional aids:

Use proper teaching aids to help the biology teacher to clarify structural and functional aspects of the topic. It helps to motivate

pupils and learn effectively. Teaching aids should be used skillfully in the appropriate time. Instructional aids may be diagrams, specimen's charts, 3-dmodels, films etc.

5. Introduction of the lesson:

Introduction of the lesson prepares the student for learning.

During introduction, the teacher can

- Review or test the previous knowledge of the students.
- Motivate the students for learning.
- Introduce new knowledge.

The lesson may be introduced effectively by presenting a demonstration, using teaching aids, asking thought provoking questions or relating a story briefly. The part is mainly concern with attention of the pupils. So the time taken for introducing the lesson should be in 3 minutes.

6. Development of the lesson:

This is the body of the lesson. This is concerned with the development of the learning activities. This step is related to the content, teaching method, teaching-aids and ability of the students. Content is the base on which learning activities are learned. So the teacher has to select teaching concepts and organize in order. Based on this content the learning activities should be developed. Based on the learning activities teaching method and teaching aids should be selected for easy acquisition of concept. The teacher has to bear in mind the age, maturity and interest of the pupils. To make sure of effective teaching process, evaluation may be carried out at every stage.

7. Recapitulation:

Recapitulation helps in recalling the lesson learned and to find out extend to which teaching is effective. Recapitulation may be done through questioning of written exam.

8. Home assignment:

Home assignment in learner to study and apply the acquired knowledge. Home assignment includes diagrams. Written materials, collection of related articles, maintaining album etc.

9. Lesson plan format:

There are five columns provided in the format lesson plan. The first column "steps" includes the motivation, lesson development, Recapitulation and assignment.

In the second column "specification" the instructional objectives are written in the form of specifications.

The third column is "learning experience" here the details regarding how the lesson taught. What is interaction between the teacher and the pupils? What are the teaching aids used, how and when the teaching aids are used etc. are written.

The fourth column is "evaluation" column. Here the questions to be put after teaching each and every concept are written.

The last column denotes the "remarks" this for the supervisor to write down the remarks if necessary.

3.3 LEVELS OF TEACHING- MEMORY, UNDERSTANDING & REFLECTIVE LEVEL

Teaching and learning process is closely related to each other and regarded as one single concept. On classroom same content is taught at different levels because it has its own structure. The learning conditions and learning objectives fall on the continuum.

So teaching learning situations may be classified according to the place where they fall on the continuum, which ranges from thoughtless to most thoughtful mode of operations.

Morris L. Biggie has classified teaching learning process into three levels which are as follows:

1. Memory level
2. Understanding level
3. Reflective level

3.3.1 Memory level

It is a memory level of teaching in which the teaching is done to provide textbook knowledge to students and the emphasis is more given to memorization rather than learning. It is not said to be the best level of teaching as memorizing anything is not good and importance should be given to understanding the concept. Well, this level helps in increasing the memory power.

Advantages of Memory Level:

- ❖ It is beneficial for young students
- ❖ Useful for slow learners

- ❖ Helpful to children for learning a new concept
- ❖ The basis for understanding the reflective level of teaching

Disadvantages of Memory Level:

- ❖ Not beneficial for higher classes
- ❖ Dominance of teacher
- ❖ Loss of recall and retention
- ❖ Interaction level is low in the classroom
- ❖ The problem of classroom management
- ❖ Students don't get a chance of self-learning

3.3.2 Understanding Level

Understanding level teaching is a more deliberate teaching method in which students interact with concepts and, as the name implies, comprehend the content. They evaluate the concepts and bring together the facts and other notions associated with them based on the students' prior awareness and comprehension level. Teachers and students are also equally interested in this phase, and in this teaching stage, teachers use debates, seminars, explanations, and other similar methods. The classroom atmosphere is welcoming and stimulating.

Advantages of Understanding Level:

- ❖ Effective learning
- ❖ Development of different cognition learnings
- ❖ Sets stages for entering into the reflective level of teaching
- ❖ The interaction level is good

Disadvantages of Understanding Level:

- ❖ Ignore higher cognitive abilities
- ❖ Less emphasis on intrinsic motivation
- ❖ No individual learning
- ❖ Teacher centered

3.3.3 Reflective level

The introspective level of teaching is also known as the reflective level. This is the most advanced level of the teaching-

learning method. This is because teaching does not end until the students have grasped the idea. It goes beyond comprehension and requires students to focus on what they have heard or understood.

This is a more advanced stage in which the learner advances one step further through the concept to comprehend the concept's multiple dimensions. This is the level of instruction in which the instructor encourages and stimulates students to think about and reflect on the content and concepts they have learned.

Advantages of Reflective Level:

- ❖ Thoughtful mode of operation
- ❖ Learner-centred
- ❖ Development of problem-solving skill
- ❖ Maximum flexibility
- ❖ Development of creativity
- ❖ Self-motivation

Disadvantages of Reflective Level:

- ❖ Not suitable for lower classes
- ❖ Time-consuming process
- ❖ Not suitable for Indian classroom
- ❖ Not applicable for dull students
- ❖ The burden on the teachers

3.3.4 Comparison of Memory level, Understanding level and Reflective Level

Memory Level of Teaching	Understanding Level of Teaching	Reflective Level of Teaching
Professor Herbert	Morrison	Biggie
Model of Memory Level of Teaching Professor Herbert originated model of teaching having five important aspects which are: focus, syntax, social system, support system and evaluation system.	Models of Understanding Level of Teaching It is known as Morrison's model of teaching. According to him, the paradigm of teaching is as follows:	Models of Reflective Level of Teaching
Focus: ❖ At this level aims of teaching are as follows: ❖ To train the mental discipline of the student. ❖ To acquire factual information. ❖ To retain the learnt material for longer time. ❖ To recall or to produce and recognize the learnt material.	Focus: ❖ The main focus of this type of teaching is to have the mastery over the concept. This model is idea-centered. It provides substantial knowledge.	Focus: ❖ To develop the ability of problem solving. ❖ To develop the critical and creative thinking among students. ❖ To enable the students for original thinking.
Syntax: The structure of the model is developed with the help of five formal steps which are known as paradigm of teaching. These are as follows:	Syntax: The structure of understanding level involves 5 steps which are as follows:	Syntax: It is based on the nature of teaching and at this step problems are solved. Here problem may be of two types: ❖ Dewey's problematic situation ❖ Kurt Lewin's problematic situation
Preparation: Important function of this step is to bring into consciousness, relevant ideas of the subject matter. The teacher plans about presentation of the contents.	Exploration: ❖ It involves testing, questioning, to explore initial learning of the students. It provides apperceptive sequence for teacher to start the chapter.	Dewey's Problematic Situation It is related to personal problem, which occurs in two situations: 1.

Memory Level of Teaching	Understanding Level of Teaching	Reflective Level of Teaching
Presentation: At this step, new ideas or new knowledge of the content is imparted in such a way that it may be linked with earlier knowledge of the students. Comparison: In this level, teaching attempts to compare old facts and new facts, its similarities and dissimilarities. Generalisation: In this step, teacher generates the situation so that students may be able to indicate the similarity of elements of new ideas and knowledge. The rhythm and recitation are used at this level. Application: At this level, teacher creates the situation so that students may use the memorized facts for the further facts. The mathematical tables are used in multiplication and division.	❖ It intends to assist teacher to arrange subject matter into psychological sequence. ❖ The role of exploration is orientation of teacher to decide how a new unit should be presented. Presentation: It involves 3 activities: ❖ The teacher presents the new content into small units and attempts to maintain continuous rapport with the students in the class. ❖ The teacher tests the student to diagnose how many students have grasped the presented content. ❖ If necessary the presentation is repeated, the teacher should not proceed toward the new unit until most of the students fully understand. Thus, at this stage teacher is more active. Assimilation: ❖ The purpose of this step is to attain thorough knowledge of the subject matter. ❖ At this step both student and teacher are active but students become more active, they do lot of work in library, laboratory, field work, assignments and also consult source materials. ❖ Students work at this step is highly individualized.	1. When a student tries to achieve his goal and he finds obstacles in his path and getting on alternative path to overcome the obstacles, thus a personal problematic situation is evolved before the learner. Here he will learn by solving it using his own insight. It is known as non-path situation. 2. When a student has two equally alternative goals. He has problem to choose one either of the two goals. This type of situation creates tension in the learner and personal problematic situation is generated before the learner, the student makes effort to find out the solution by using his original thinking and ability. Kurtlewin's Problematic Situation According to Kurtlewin, every learner has his own goal and this goal controls his behaviour. He has a field and his life's space where his nature is psychological and social. The situation of both creates tension in the minds of the students. In reflective teaching the main task of a teacher is to create a problematic situation before the learner and he has to formulate tentative solutions for the problem.

Memory Level of Teaching	Understanding Level of Teaching	Reflective Level of Teaching
	❖ A mastery test is also administered at the end of assimilation. If students cannot pass the test then they are asked for further assimilation. Organisation: ❖ This step provides an apperceptive sequence when students' have passed mastery test then they move towards the organisation step. Following observations are included in this step- ❖ In this step the student determines what content has to be written. Morrison states that "the student receives mastery when he is able to write down the presented content and the assimilated data clearly". ❖ This step is specially essential in subjects of extensive content which include a large no. of elements in one learning unit. Recitation Following activities are included in this step: ❖ Each student presents the content orally before the teacher. ❖ Morrison plans daily recitation for this step. ❖ Recitation may take the form of written paper.	Here teacher has to encourage the learner to have more than one hypothesis for the problems, after that the verification of the hypothesis. On the basis of testing hypothesis, conclusions are drawn. These conclusions are the original thinking of the learner.

Memory Level of Teaching	Understanding Level of Teaching	Reflective Level of Teaching
Support system: This system requires some supportive devices to make learning more effective. More meaning full material should be used to present at this level the subject matter should be made definite in its structure. The audio visual aids may be used to involve the students. The content should be specified and present in an observable form.	Support system: This step is helpful to make this model more effective so that goals may be achieved. The teaching aids should be used at presentation stage. It may not be repeated several times, the audio-visual aids and other devices may be used for an effective presentation. This system is much more useful at assimilation stage. The subject matter can be gathered from various sources. The programmed instruction material may be provided to the learner because this stage is highly individualized. The learners should be given an opportunity to practice the learnt materials according to their own efforts.	Support system: This step is much important to make the teaching more effective. The learner should be helped by social awareness and social values to feel the need of the problem. He should be encouraged to formulate as many hypothesis as possible. This is only possible by the personal involvement and high level of aspiration. The learner students should be oriented to collect the evidences for the verification of the hypothesis. This will lead in drawing the conclusion regarding the problem.
Social system: This system consists of authoritarian behaviour. The teacher is more active and his main act is to construct and present the subject matter systematically and logically in the classroom. The learners are passive listener only and they have to memorize and reproduce the content and their activities are directed by the teacher. At this level teacher provides motivation through verbal praise and punishment, if the learner is motivated or reinforced continuously then they learn better.	Social system: At this step teacher should be more active to see that he should not be authoritarian. He has to act as a leader of the group and he has to guide and motivate the students only. Their effort should discover the facts. Teacher's behaviour should be flexible, dynamic and full of human touch. A teacher should have full faith in the learners. The source of motivation at this level of teaching is not only extrinsic but also intrinsic and personal involvement of teacher is also necessary. The teacher's main act is to create need and interest among students to acquire understanding of new concepts. Here teacher can raise the level of aspiration of students as well as class as a whole.	Social system: The teacher should be creative, tactful and insightful and should be able to create a problematic situation. The role of the teacher is to assist the learners and make them active to evolve the problem, the learner should also be very active to find out the solution of the problem by objective and empirical procedure, the self-motivation and level of aspiration are the sources of motivation at this level.

Memory Level of Teaching	Understanding Level of Teaching	Reflective Level of Teaching
Evaluation system: It is important aspect of teaching because it produces evidences about the realization of goals of teaching. Usually oral tests are given to measure the retention of learnt material. The objective tests of recall and recognition type are administered to evaluate the knowledge of the students.	Evaluation system: The organisation and recitation provides evidence about the effectiveness of this model. The essay and objective type tests are used for evaluating the results of this model. The multiple choice and analogy type item are best items for evaluating the understanding of the students.	Evaluation system: The workability of the teaching is difficult to measure, the effect of teaching at this level, the written test should be used to measure higher level of cognition. The essay type tests are the test devices to evaluate the result of reflective level of teaching. The other techniques should be used to measure the environment in the problem, attitudes, adjustment and critical thinking of the learner.
Suggestions for Memory Level Teaching ❖ Frequent recall or reproduction of content should be practiced. ❖ The fatigue factor should be avoided in the presentation. ❖ Whole to part method should be used. ❖ The content should have definite structure. ❖ Teaching content should be integrated. ❖ Contents should be presented systematically. ❖ Continuous reinforcement should be provided to the students. ❖ Students' retention can be increased by practice.	Suggestions for Understanding level of Teaching ❖ Teacher should move towards understanding level of teaching when the student pass out memory level. ❖ The teacher should give the test of the presentation to know the next step of assimilation, after this he should move-towards organisation or recitation step. In other words, these steps should be followed systematically. ❖ Teacher should have personal environment in teaching and the subject to be taught. ❖ Teacher should also provide psychological motivation to the students and raise their level of aspiration. ❖ Teacher should identify the problems of organisation and presentation and provide solutions of the problems.	Suggestions for Reflective level of Teaching ❖ It should be only recognized when the learners have passed the memory and understanding level test. ❖ The learner should be sensitive for the problem and teacher should create the need of the problem. ❖ Teacher should raise the level of aspiration of students for the involvement in the problem. ❖ Teacher should create problematic situation which may develop the original and critical thinking of the learner. ❖ The student should feel about the problem so that they may formulate the hypothesis of the problem. ❖ The learner should be objective in collecting the evidences for the problem. ❖ The learner should be given freedom to think independently.

3.4 UNIT PLAN

3.4.1 Unit: (Preston)

He defines a unit as "a large block of related subject matter as can be over-viewed by a learner".

A course of study can be divided into units. A unit is defined as the series of related learning experiences built around one central topic. A unit develops a generalization through the learning of related facts, concepts and views (or) viewpoints. Therefore a unit is a large segment of subject matter having a common theme/idea.

3.4.2 Unit Planning:

Unit planning is a part of year planning and it is a middle ground between course / year planning and lesson planning. It is longer than lesson plan, but shorter than course planning. There are two types of units.

1. Resource Units

2. Teaching Units

The Resource unit is intended to be a general guide in assisting the teacher to enrich the teaching unit. It is more comprehensive, than the teaching unit. It is according to the level of the knowledge of the teacher. It requires pedagogical skill for presenting Resource units are developed by the curricular experts and classroom teachers.

The teaching unit is important and useful to the prospective or In-service teachers.

A unit plan is basically a list or series of lesson plans designed around a specific topic, project, content, etc, and among other reasons that apply for planning in general, it is important for teachers to design one because a great deal of learning is directly correlated to teachers' planning and preparation.

3.4.3 Designing a Unit Plan

Unit plans are consistent with State curriculum guidelines. Therefore, you should be aware of the standards as well as the contents for the subject matter you intend to write about in a unit.

- ❖ You must stick to a timeline to complete a unit.
- ❖ You should take some time to review what resources are already available to you (for designing your Unit Plans you may unquestionably resort to your Curriculum Specifications and the textbooks you receive).

- ❖ You should know about your students' needs and interests (through a diagnosis and/or survey) as well as have pedagogical resources available before you begin planning the lessons you choose.

- ❖ Once you review what resources you can resort to, you should create a diversity of assessment tools to evaluate learning because you should ensure students meet the objectives through evaluation. both formative and summative

- ❖ After considering your students' needs and the timeline, select what may fit their learning styles and use lessons and activities that motivate their curiosity and interest.

- ❖ You can follow a standard Procedure to prepare a unit.

The Procedure may begin with the objectives but also include the competences addressed, materials, lessons, assessment, resources, etc.

3.4.4 Components in a Unit Plan

- ❖ **Unit Title & Subject Area:** The theme of your unit and subject you are teaching.
- ❖ **Communicative competence components:** Briefly list the ones covered in this unit plan.
- ❖ **Classroom/ Level:** The classroom in which the subject is taught and the level according to the Curriculum Guidelines.
- ❖ **Unit Rationale:** Your rationale is your overall justification for the unit, and basically what you will have to do is to provide a paragraph explanation to describe the value of the unit that is presented, the why the unit is important, how it will benefit students and where it will fit in the curriculum. This is prepared by individual teachers based on the students, school, community, and on curricular and instructional objectives and needs.
- ❖ **Goals:** They are also called long-term objectives and describe the purposes or major concepts that will be taught. It is expected that from each goal, several specific objectives could be pulled out. The goals will cover the entire unit whether it is 3 lessons or 8.
- ❖ **Objectives:** These come from the goals and represent the content areas or skills that would be included on all of the lesson plans for the unit. They are more specific than the goals. Objectives must be measurable. (Short-term or immediate objectives).

- ❖ **Content:** Identify and organize the topics that are included in the unit.
- ❖ **Learning activities:** The main methods/activities for teaching each goal. While not as detailed as the instructional procedures in a lesson plan, they should contain all key information. Example: display a power point presentation, introduce the Syntax for writing a programme.
- ❖ **List of Sources:** All sources consult in preparation of the unit should be listed. Remember multiple sources are expected.
- ❖ **List of Materials:** List all materials needed to teach the unit that are not readily available to the teacher every day.
- ❖ **Assessment:** Identify the ways students learning will be measured as well as how students will receive feedback and list the strategies and instruments you will use. (remember the way you measure students' learning will depend on what you want them to learn-i.e. objectives and the learning activities involved)
- ❖ **Explanation of how the unit plan addresses a particular student profile.** Demonstrate how your plan addresses the interests and needs of your students (you may resort to the results you got from the administration of diagnostic tests at the beginning of the school year because that may have provided you with some information on your learners' strengths and weaknesses). The example for unit planning is given below.

3.4.5 Topic: Fundamentals of Internet Computing

Objectives: Objectives must be written clearly explained. For an example, if we are going to teach the unit Fundamentals of Internet Computing, we can consider objectives as follows.

- On successful completion of this unit students should be able to:
1. Understand the fundamental principles of computer systems and the Internet;
 2. Design web sites;
 3. Use simple client-side programming;
 4. Understand the principles of user interface design and human-computer interfaces.

Learning Experiences: This module provides an introduction to the key concepts of Internet computing. Starting with the fundamentals of computer systems and the Internet, students' progress to learn how to design web sites and how to utilize

simple client-side programming. Issues related to user interface design and human-computer interfacing (HCI) are covered. Broader issues related to the use of the Internet for Blogging and Social Networks are discussed. The practical element of the module allows students to develop skills necessary for web site design using simple client side programming.

Method & Materials

1. **Method:** Inductive, deductive, analytic, synthetic, Problem solving etc
2. **Techniques:** Drilling, Questioning, assignments
3. **Materials:** CAI, Chart, Model, Chart, Real Objects.

Evaluation

Formal Written Examination, Continuous Assessment

Feedback

The following can be given - Effective Computer Package

- ❖ Reteach
- ❖ Drilling
- ❖ Remedial teaching

3.4.6 Designing a unit plan

A unit plan is detailed outline for series of interrelated lessons on a selected topic of study which lasts for two to four weeks. Unit plan is an organized instructional material which teachers develop and select from the available sources to impact instruction.

Each unit is analyzed and the general instructional objectives are stated. Each general instructional objective is further analyzed and stated as specific instructional objective in terms of terminal behavior of the learner. Then teaching points are prepared. The teaching methods are identified. Teaching aids are selected. Methods of evaluation are decided. Resources materials for further learning like references books films, film strips etc are listed. Time needed for each unit is estimated and allocated.

So, unit plan considers the following aspects.

- a) "Why" aspect-deciding and stating instructional objectives.
- b) "What" aspect-analysis of the unit.
- c) "how" aspect-method and learning experiences
- d) "Evidence for achievement"-evaluation.

FORMAT

Subject:

Unit:

class:

General instructional objectives,

- 1.
- 2.
- 3.

S.No	Topics	No. of lessons topics	Time required	Specific instructional objectives	Teaching method	Teaching aids
1.						
2.						
3.						
4.						
5.						

3.4.7 Advantage of unit plan

- ❖ An unit plan should breakup a lengthy unit into smaller sub units so that pupils can easily grasp the ideas.
- ❖ Without losing the continuity, a teacher can present a lesson, principles and follow a systematic teaching.
- ❖ It helps the teacher to plan a variety of learning experiences keeping in mind the individual differences, the nature of content and objectives to be achieved.
- ❖ The pupil can see various facts, principles and can able to relate them with the unit.
- ❖ Unit planning has a proper provision for the diagnosis of the learning difficulties of the students and subsequent remedial instruction.
- ❖ Unit planning paves the way for a proper and appropriate daily lesson planning.

3.5 YEAR PLAN

A course of study can be divided into units. A unit is defined as the series of related learning experiences built around one central topic. A unit develops a generalization through the learning of related facts, concepts and views (or) viewpoints. Therefore a unit is a large segment of subject matter having a common theme/idea.

Lesson plan should give clear indication what students are going to do for learning and what they are expected to grasp in the class room during the process of teaching

Planning begins with the goals of teaching-learning situation, carried through planned means of achieving these goals and ends with plans for evaluating the efforts of all persons involved.

So plans should be made in terms of goals to be achieved. Therefore, the stages in planning should be

1. Course planning which could determine the general objectives, specific objectives, and accordingly select the course to be taught.
2. Organization of the selected material into meaningful segments of activity and experience.
3. Plan the daily work around a device known as 'Lesson Plan' In this lesson you are exposed to the various components of a lesson plan in Computer Science.

An annual plan or Year plan is intended to provide teachers and school administrators with an overview of curriculum guidelines at a glance, and therefore it can definitely help educators to stay on track each month, for instance in meeting the curriculum goals that have been set up for the school year. Year planning is done. By a teacher at the beginning of the year for the entire course.

3.5.1 Designing Annual Teaching Plan

The following are only suggestions to the layout of your annual plan, and therefore, you may feel free to adapt, modify or even improve any format resulting. From them to fit your specific needs and the needs of your group(s) of learners.

Section One

Start by labeling your plan, and providing general information about the subject you are designing the plan for, school's name, school year, and target group (s) you will be teaching in the first section.

1. General Information

Area: C – Programming: Class Higher Secondary
 School Year: 2015 – 2016 Target group (s): XI Standard

Section Two

Take your Curriculum Guidelines to look at the objectives per school year that have been proposed for each component of the Programming competence in C.

2. Objectives

On successful completion of this course the students should be able to

- Understand the basics of C language
- Use conditional statements and loops
- Write programmes using of arrays
- Use functions for writing programmes
- Use pointers for writing programmes
- Use the skill of file processing

Section Three

3. Model Annual Plan

Insert a blank table for your template in a word processing document and list the months that correspond the present school year (ie June to April) in the first column

Month/weeks	I	II	III	IV
June				
July				
August				
September				
October				
November				
December				
January				
February				
March				
April				

Section Four

4. Methodological Strategies

Read the Curriculum specifications you have been provided with and summarize information on the methodological strategies as well as the genres and text types that have been proposed as

tools for developing each programming skill with your corresponding groups of learners.

Unquestionably, you may feel free to propose and write down any other methodological strategy (ie) you consider pertinent by keeping in mind they should reflect the basic principles Computer Programming:

- ❖ Make real procedure of programming.
- ❖ Provide opportunities for learners to experiment and try out what they know.
- ❖ Be tolerant of learners' errors as they indicate that the learner is building up his or her Programming Competence.
- ❖ Provide opportunities for learners to develop both accuracy and fluency.
- ❖ Link the different skills such as analyzing problem, preparing flow chart, writing algorithms, writing Programmes, debugging the programmes and executing the programmes together, since Situation they usually occur in the real situations.
- ❖ Let students induce or discover Programming rules.

Section Five

5. Resources and Equipment's,

Briefly list the didactic resources (ie, official textbooks as well as supplementary materials and equipment) that are available at your institution or you have made or provided and which you consider are appropriate to facilitate the students' learning process.

Section Six

6. Evaluation

Take the Curriculum Guidelines document and carefully look at the assessment indicators per skill for your target group. Then describe briefly the type of evaluation you will carry out with your students (ie, diagnostic, formative, and/ or summative) according to State regulations and when it will be done (at the beginning, middle or end of the first second term, school year, etc.), as well as the type of assessment you will carry out (ie, formal/ informal) and the type of assessment activities and / or

instruments you will use (e.g. quizzes presentations, writing Programmes etc.)

3.5.2 Characteristics of Year Plan

- ❖ The number of units to be covered as per the syllabus during the academic year.
- ❖ The number of periods needed to cover each unit, the number of periods allotted for Computer Science per week.
- ❖ The number of working days per term and for the year.
- ❖ The number of days allotted for project work, revision tests & examinations.
- ❖ The number of holidays during the academic year.

3.5.3 Advantages of Year Plan

Year planning helps the teacher in planning for the entire course for the whole of the academic year.

- ❖ It keeps the teacher on the right track.
- ❖ It enables the teacher to cover the syllabus within the allotted time planning saves waste of time and energy.
- ❖ It helps the teacher in orderly and systematic teaching.
- ❖ It gives a lot of self-confidence to the teacher as she is sure of what is expected of her during each period/week/month /term etc.
- ❖ It will help to carry out all the curricular and Co-curricular activities in time without haste and anxiety.
- ❖ It helps in allotting time for lab, projects, revision, tests and examinations etc.

3.6 PREPARATION OF A MODEL LESSON PLAN Instructional/General Objectives for Computer Science Teaching

The specific objectives under cognitive, affective and psychomotor domain are clubbed into three, two and one respectively for the purpose of formulating general instructional objectives for science teaching. These are as follows.

1. Knowledge
2. Understanding and
3. Application are from the cognitive domain,
4. Interest and
5. Attitude are from the affective domain and
6. Skills are from the psychomotor domain.

By now you are familiar with these terms. Let us now discuss these objectives as they are used for instruction purpose along with expected change in behaviour of learners.

Knowledge

Knowledge as you know is the lowest level of cognitive ability. Therefore, the first objective of teaching science is related to acquisition of knowledge by the students. It states:

“Students will acquire the knowledge of facts, concepts, principles, processes and techniques etc”.

The behavior specification to demonstrate the achievement of this objective is that pupil will be able to:

- ❖ Recall terms, facts, concepts, principles and processes etc.
- ❖ Recognize facts, terms, concepts, principles and processes etc.

You know that words like ‘recall’ and ‘recognize’ are the action words which represent the behavior of individuals. This overt behavior can be observed and measured by teacher as well as by an evaluator or any observer.

Understanding

The second objective of science teaching is related to the next higher cognitive ability i.e. comprehension or understanding. It states:

“To develop the ability to understand facts, concepts, principles, theories and techniques etc”.

Behavior specification of objectives are that pupils:

- ❖ Illustrate terms, facts, concepts, principles etc. by citing examples.
- ❖ Express the same fact or concept in different ways by way of explanation.
- ❖ Locate errors in known situation and correct them.
- ❖ Compare and contrast between related terms and concepts.
- ❖ Classify objects, facts or any information.
- ❖ Discriminate between allied substances or concepts.
- ❖ Identify relationship between various facts and concepts.

- ❖ Extrapolate for known information.
- ❖ Translate symbolic statement into verbal statement and vice versa.
- ❖ Interpret data, charts, graphs etc.
- ❖ Detect errors in faculty statements.
- ❖ Verify facts.
- ❖ Solve numerical problems.

Application

The essence of teaching/learning, as you know, is the application of knowledge. You perform various activities or solve many problems in our daily life. You are able to do this only because you have the knowledge and understanding of the things around yourself. The third objective aims at using knowledge in new situations. It states:

"To develop the ability to apply knowledge of concepts and principles of science in new or unfamiliar situations:

Behaviour specification of the objectives are that the pupil will:

- ❖ Analyze situation or problem
- ❖ Formulate hypothesis on the basis of observations
- ❖ Select appropriate methods and material for testing the hypothesis
- ❖ Give reasons for happenings
- ❖ Draw inferences, conclusions and generalizations
- ❖ Predict results or happenings on the basis of known facts

Skills

We know that 'Science learning is science doing', for learning science. Various skills are required for performing activities and acquiring knowledge. Also, type of knowledge acquired by an individual depends upon the processes applied for acquiring it. The fourth objective of teaching science states:

"To develop observational, experimental, manipulative and drawing skills".

Cognitive Behavioral Terms for Stating Specific Learning Outcomes

KNOWLEDGE	COMPREHENSION	APPLICATION	ANALYSIS	SYNTHESIS	EVALUATION
Defines	Converts	Changes	Breaks down	Categories	Appraises
Describes	Defends	Computes	Diagrams	Combines	Compares
Identifies	Distinguishes	Demonstrates	Differentiates	Complies	Concludes
Labels	Estimates	Discovers	Discriminates	Compose	Contrasts
Lists	Explains	Manipulates	Distinguishes	Creates	Criticizes
Matches	Extends	Modifies	Identifies	Devises	Describes
Names	Describes	Operates	Illustrates	Designs	Discriminates
Outlines	Generalize	Predicts	Infers	Explains	Justifies
Reproduces	Discriminates	Prepares	Outlines	Generates	Interprets
Paraphrases	Gives examples	Points out	Modifies	States	Relates
Predicts	Infers	Relates	Organizes	Recalls	Supports
Rewrites	Produces	Selects	Plans	Recognises	Summarizes
Solves	Relates	Rearranges	Reconstructs	Revises	
	Shows	Uses	Subdivides	Reorganizes	
	Separates			Rewrites	
				Summarises	
				Tells	
				Writes	

AFFECTIVE DOMAIN

Receives	Responds	Values	Organises	Characterizes
Asks	Answers	Accepts	Advises	Accepts
Accepts	Assists	Attains	Associates	Changes
Attends	Completes	Chooses	Changes	Characterizes
Is beware of	Derives	Completes	Compares	Decides
Catches	Discusses	Decides	Completes	Discriminates
Discovers	Develops	Develops	Co-ordinates	Demonstrates
Experiments	Helps	Demonstrates	Correlates	Experiment
Follows	Labels	Increases	Determines	Faces
Identifies	Lists	Influences	Generalizes	Identifies
Observes	Presents	Influences	Integrates	Proves
Prefers	Practices	Participates	Judges	Revises
Perceives	Records	Prefers	Organizes	Solves
Receives	Selects	Recognizes	Projects	Verifies
Selects	States		Synthesizes	
Verifies	Writes			

PSYCHOMOTOR DOMAIN

Perceives	Imitates	Manipulates	Precision	Articulates	Naturalizes
Balances	Repeats	Draws	Makes	Adapts to complex situation,	Reorganizes the automatic performances
Catches	Reproduces	Experiments	Involves		
Discovers	Imitates	Operates	Performs with accuracy		
Explores		Manipulates	Performs sequentially		
Identifiers		Develops skill			
Touches					
Hears					
Memorizes					
Sees					
Smells					
Traces					

The objectives of teaching science have been changing from time to time. The rapid progress of science and technology in recent years has changed the objectives of science in schools. The stress from mere knowledge of facts was shifted to the development of concepts, abilities, skills, and appreciation etc. hence the achievement of all these aims should be the challenges of Science teachers.

3.7 DESIGNING A LESSON PLAN

3.7.1. Model Lesson Plan No 1 - Commerce and Accountancy

Name of the Student Teacher :
 Name of the Guide Teacher :
 Name of the School :
 Standard : XI STD
 Subject : Commerce
 Topic : Classification of Economic Activities.

Date :
 Time :
 Instructional Objectives: The Pupil

1. Explain the meaning of human activity and its types
2. Specify the term of economic activity.
3. Arrange the economic activities and their characteristics.
4. Look over the functions of economic activity.
5. List out the types of business activities

Instructional resources required:
 1. Chart showing Meaning of human activity and definition of economic activity.

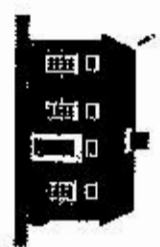
2. Flow chart showing types of economic activity.
 3. Flash Cards - functions of economic activity.
- Previous Knowledge of Learners
1. What is purchase & sale of goods is called?
 2. What is the outcome of business?
 3. What is the source of initiation to economic activity?

Content	Specification	Learning Experiences	Evaluation
Meaning of Human Activity Human activity is an activity performed by a human, being to meet their needs and wants or they be for personal satisfaction.	Explain	Teacher explains the meaning of human activity and its types using chart.	Human activity is performed by human being to meet



It can be categorized into economic and non-economic activities.		Students Observed it	
Definition of economic activity Economic activities are those activities which are undertaken to earn money or financial gain for livelihood.	Specify	Teacher defines economic activity using chart 	Economic activities are undertaken to earn _____
Types of economic activities and their characteristics Generally, people engage themselves in such activities on a regular basis and are said to be engaged in their occupation. Occupations may be classified into three categories based on the following: A. Employment B. Profession C. Business A. Employment It refers to the occupation. The one who is employed by others are called employees and the one who employs others is called employer.	Arrange	A flow chart showing the arrangement of economic activity. 	How many occupations are there in economic activity?
		Students observe The flow chart.	Define Employment. Occupation of a Doctor is

Example : Clerks, Bank officials, Factory workers etc.,		
Characteristics Remuneration is given to employees for delivering value of services. An employee must abide by the rules and regulations of the organization.		
B. Profession Professions are those occupations which involve rendering of personal services of a special and expert nature. A profession is something which is more than a job. Example: Lawyer, Doctor, Engineer...		
Characteristics Service motive should be upmost in the minds of people rendering professional service.		
C. Business Business may be defined as "an activity involving regular production or purchase of goods and services for sale, transfer and exchange with an object of earning profit."		

Business activities are connected with raising, producing or processing of goods.			
Functions of economic activity	Look over	The teacher explains about the economic activity.	Main motive is _____
Motive: Sole motive is to earn money or financial gain. e.g., Working as a lawyer.	Using Flash Card 		
Money Measurement: All economic activities can be valued in monetary terms. e.g., Doctor charges ₹500 as consultation fee			Give an example for money measurement
Relationship: Economic activities are related to creation of wealth. e.g., Ram saved part of his salary to purchase a house of his own.			
Outcome: All economic activities result in production, procurement, distribution and consumption of goods and services.			

c.g., Nokia produces cell and sells across India through its distributors.			
Duration: Economic activities are repetitive. They are done on a regular basis to earn a living. e.g., Kufli ice cream seller sells ice creams every evening.			
Source of Initiation: Economic activities are initiated to satisfy human needs and wants			
Types of business activities Business activities are classified on the basis of size, ownership and function.	List out	Students understand the activities.	
1. Activities on the Basis of Size a. Small Scale - It require less capital. They employ small number of workers and produce the goods. Example, handlooms		Teachers explain the types of business activities using chart. 	How many types are there in activity based on the size?
		Students understand the types of business activities.	Expansion of PWD

b. Large Scale - It require huge capital. They employ large number of workers and produce the goods on large scale. Example, Textile mill.		Give an example for industry.
2. Activities on the Basis of Ownership a. Private Enterprises - An enterprise is said to be a private enterprise where it is owned, managed and controlled by persons other than Government. Example, b. Public Enterprises - An enterprise is said to be a public enterprise where it is owned, managed and controlled by Government or any of its agencies or both. (i) Departmental undertaking Public Works Department (PWD) (ii) Public Corporation - Oil and Natural Gas Corporation (ONGC) (iii) Government Company - State Trading Corporation (STC)		

c. Joint Enterprises - It is owned, managed and controlled by Government and private entrepreneurs. Example - Maruti Suzuki			
3. Activities on the Basis of Function Business activities may be broadly grouped into two categories. a. Industry - It includes all those business activities - producing or processing of consumer goods. Example - bread, butter, cheese, shoes, or capital goods like machinery. b. Commerce - It establishes a link between the producers and consumers of goods and maintains a smooth and uninterrupted flow of goods from producers to consumers.			
Follow up Activities 1. List out the types of business activities? 2. Describe the functions of economics activities			
Signature of the Guide Teacher			Signature of the Student Teacher